Predictive Model of Problematic Internet Use Based on Physical Biomarkers

Deliverable I: Description of the Problem, Data Analysis, and State of the Art

Xiomara Betancur Isaza
Facultad de Ingeniería
Universidad de Antioquia
Medellín, Colombia
xiomara.betancuri@udea.edu.co

Oscar Albeiro Blandón Agudelo
Facultad de Ingeniería
Universidad de Antioquia
Medellín, Colombia
oscar.blandon@udea.edu.co

***Abstract*—Problematic Internet Use (PIU) in childhood and adolescence is associated with sleep disorders, depressive symptoms, anxiety, stress, and poorer academic performance. This first deliverable describes a supervised machine learning problem aimed at predicting the Severity Impairment Index (*sii*), a four-level ordinal index of functional impairment due to internet use, using physical biomarkers, physiological measurements, and contextual data from the Healthy Brain Network (HBN) dataset distributed by the Child Mind Institute on Kaggle. The labeled sample contains 2,736 participants and is severely imbalanced (the severe class represents only 1.24%). We characterize the dataset, the systematic missing-data patterns, and the leakage-control rules (exclusion of the PCIAT questionnaire and the identifier). We also define the ordinal learning setup, the Quadratic Weighted Kappa (QWK) as the main metric, the Macro F1-score as a complementary metric, and a 5-fold stratified cross-validation scheme. A review of the state of the art on tree ensembles, multimodal architectures, and neural networks reveals three gaps: absence of rigorous statistical validation, risk of information leakage in imputation, and poor treatment of the severe class. Our proposal addresses them by comparing five families of algorithms with imputation and scaling encapsulated within each fold, and by evaluating dimensionality reduction with PCA and UMAP.**

***Index Terms*—Problematic internet use, machine learning, Healthy Brain Network, ordinal classification, quadratic weighted kappa, class imbalance, data leakage, physical biomarkers.**

I. Problem Description

A. Context and Utility of a Machine Learning-based Solution

Problematic Internet Use (PIU) during childhood and adolescence constitutes a phenomenon of great clinical and social interest due to its profound implications on the daily functioning and psychosocial well-being of minors. The scientific literature has rigorously documented the associations between compulsive and uncontrolled consumption of digital devices and various mental health disorders, highlighting a strong link with sleep disorders, depressive symptoms, clinical anxiety, stress, and a marked deterioration in academic performance [1]. Likewise, early detection tools have been developed that demonstrate the feasibility of building efficient screening mechanisms based on objective and observable variables in the youth population [2], [3].

In this context, the present project uses data from the Healthy Brain Network (HBN) initiative, developed by the Child Mind Institute (CMI) and distributed through Kaggle [4], with the aim of predicting the target variable Severity Impairment Index (*sii*), a four-level index that quantifies the severity of functional impairment due to internet use.

The value of a Machine Learning (ML)-based solution does not lie in replacing specialized assessment or clinical practice, but in exploring whether a heterogeneous set of physical biomarkers, physiological measurements, physical activity indicators, and contextual data can identify latent nonlinear patterns associated with the different levels of severity of *sii*. In this way, the model is conceived as a predictive, auxiliary, and non-invasive early warning tool that can support mass screening programs in school and public health settings.

B. Database Composition

1) General Composition and Structure of the Dataset: The data comes from the clinical and research sample of the HBN initiative and is officially distributed on Kaggle [4]. The training dataset comprises two structured components:

- **Main structured table:** it records 3,960 participants in single rows and 82 columns that summarize demographic characteristics, psychological questionnaires, physical tests, and anthropometric evaluations.
- **Accelerometry time series:** continuous wrist actigraphy data sampled every 5 seconds over a period of up to 30 days, available for 996 participants (25.15% of the total sample). Due to its low coverage and the risk of massively multiplying the absences in the static matrix, continuous accelerometry is excluded from the initial version of the pipeline.

2) Target Variable and Severe Class Imbalance: The target variable *sii* presents two determining methodological characteristics:

- **Unlabeled samples:** the label is absent in 1,224 participants (30.91% of the dataset), so the effective sample for supervised learning is strictly composed of 2,736 labeled observations.
- **Severe class imbalance:** within the 2,736 useful observations, the distribution by severity is markedly uneven:

- Class 0 (None): 1,594 participants (58.26%)
- Class 1 (Mild): 719 participants (26.28%)
- Class 2 (Moderate): 378 participants (13.82%)
- Class 3 (Severe): 34 participants (1.24%)

The class of greatest clinical importance (Severe) represents only 1.24% of the labeled sample, establishing an imbalance ratio of approximately 47 to 1 compared to the majority class. This necessitates the incorporation of cost-weighting strategies and evaluation metrics that are not biased by class frequency.

3) Characterization of Clinical Variables and Instruments: The 82 characteristics of the dataset are operationally summarized in Table I according to the corresponding battery or assessment instrument.

4) Information Leakage Control (Data Leakage): To ensure the methodological rigor of the model, two strict variable exclusion rules are applied:

1) **Exclusion of the PCIAT questionnaire:** the 20 items of the PCIAT and their total accumulated score are categorically excluded as predictors. Since the *sii* label is derived through deterministic cuts on that score, its inclusion in the training matrix would generate direct information leakage, invalidating the model evaluation.
2) **Exclusion of the identifier:** the participant identification column (`id`) is removed due to lack of predictive value.

5) Analysis of Missing Data and Imputation Strategy: Except for age, sex, and identifier, the remaining variables show absence patterns ranging from 10.6% to 88%, with most concentrated in the 30% to 50% range. Absences occur in systematic blocks corresponding to unadministered tests. The treatment strategy includes:

- **Threshold filtering:** removal of columns with more than 70% missing data.
- **Multivariate imputation:** imputation of continuous variables using the median or `KNNImputer`, and imputation of categorical variables using the mode.
- **Absence indicators:** creation of binary indicator variables for absence by clinical block to capture information on the absence pattern.
- **Isolation by fold:** all imputation statistics (medians, modes, and KNN neighbors) are calculated exclusively on the training set of each fold of cross-validation (Stratified K-Fold CV), avoiding data contamination.

6) Encoding and Preprocessing Strategy: Preprocessing standardizes the feature space as follows:

- **Binary variables:** sex is directly encoded on a 0/1 scale.
- **Nominal categorical variables:** the enrollment season is transformed using One-Hot Encoding.
- **Ordinal variables:** the hours of technology use and the FitnessGram zones are kept as integers mapped hierarchically.
- **Continuous variables:** they are standardized with Z-Score (subtracting the mean and dividing by the standard deviation) to equalize the scale of physical and bioimpedance measurements.
- **Target variable:** the label *sii* is encoded as an ordinal integer in $\{0, 1, 2, 3\}$.

C. Paradigm and Learning Configuration

1) Supervised Learning Paradigm: The problem is formally addressed under the paradigm of Supervised Learning. The training set is defined as $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, where $\mathbf{x}_i \in \mathbb{R}^d$ represents the standardized feature vector of participant i (demographic, anthropometric, physical fitness, BIA, and sleep), and $y_i \in \{0, 1, 2, 3\}$ corresponds to the observed severity label. The objective is to learn a mapping function $f : \mathcal{X} \to \mathcal{Y}$ that minimizes the expected empirical risk on unseen data.

2) Ordinal Multiclass Classification Setup: Unlike a standard nominal classification, the *sii* variable has a strict hierarchical ordinal structure of progressive clinical severity: 0 (None), 1 (Mild), 2 (Moderate), and 3 (Severe). Two technical modeling approaches are established:

1) **Continuous regression with threshold tuning:** regressors ($\hat{y} \in \mathbb{R}$) are trained and three thresholds $\tau_1 < \tau_2 < \tau_3$ are optimized on the training set to map the continuous output to the integer scale $\{0, 1, 2, 3\}$, directly maximizing the evaluation metric.
2) **Direct multiclass classification optimized by ordinal metrics:** classifiers are trained by adjusting their hyperparameters to maximize ordinal concordance in the discrete space.

3) Main and Complementary Evaluation Metrics:

a) Main metric: Quadratic Weighted Kappa (QWK): The QWK measures the agreement between the actual and predicted labels, correcting for the agreement expected by chance and penalizing distant differences using the quadratic weight matrix:

$$W_{ij} = \frac{(i-j)^2}{(k-1)^2}, \qquad k = 4 \tag{1}$$

$$E = \frac{\mathbf{r} \otimes \mathbf{c}}{N}, \qquad E_{ij} = \text{expected frequency by chance} \tag{2}$$

$$\kappa = 1 - \frac{\sum_{i=1}^{k}\sum_{j=1}^{k} W_{ij} O_{ij}}{\sum_{i=1}^{k}\sum_{j=1}^{k} W_{ij} E_{ij}} \tag{3}$$

where O_{ij} is the observed confusion matrix, and $\mathbf{r}$ and $\mathbf{c}$ are the row and column marginals. A QWK value of 1 indicates perfect agreement, QWK $= 0$ is equivalent to performance no better than random, and QWK < 0 reflects an agreement worse than random.

b) Complementary metric: Macro F1-Score: Since class 3 (Severe) represents only 1.24% of the sample, the Macro F1 calculates the unweighted mean of the individual F1-scores of each class:

$$\text{F1-Macro} = \frac{1}{k}\sum_{i=0}^{k-1} \text{F1}_i \tag{4}$$

This metric assigns equal weight to all categories regardless of their frequency, evaluating the model's actual ability to detect severe and moderate cases.

Table I
CHARACTERIZATION OF VARIABLES AND INSTRUMENTS OF THE DATASET

Instrument / Domain	Representative Variables	Encoding and Preprocessing
Demography and Seasonality	Age, gender, enrollment season	Numerical age; gender 0/1; season with One-Hot Encoding
Self-Reported Technology Use	Daily hours on the computer or internet (0 to 3)	Ordinal, hierarchical integer
Global Functioning	Children's Global Assessment Scale (CGAS) score	Continuous numerical
Physical Measurements and Anthropometry	BMI, height, weight, waist circumference, systolic and diastolic blood pressure, heart rate	Continuous numerical, Z-Score
Physical Fitness (Treadmill)	Maximum stage reached, time in minutes and seconds	Continuous numerical, Z-Score
Physical Fitness (FitnessGram Tests)	Sit-ups, push-ups, sit-and-reach, handgrip strength	Continuous numerical and ordinal zones
Bioelectrical Impedance Analysis (BIA)	16 indicators (body fat percentage, total body water, BMR, etc.)	Continuous numerical, Z-Score
Sleep Evaluation (SDS)	Total raw score of sleep disturbances and standardized T-score	Continuous numerical
PCIAT Questionnaire	20 individual items and total accumulated internet addiction score	Excluded as predictors (data leakage)

4) Validation and Imbalance Control Strategy: The validation is based on a 5-Fold Stratified Cross-Validation (Stratified K-Fold CV, $K = 5$). Stratification preserves the exact proportion of the four *sii* classes in each fold, ensuring that the minority class (*sii* $= 3$, with 34 samples) is adequately represented in all partitions.

II. STATE OF THE ART

The development of automatic systems for the assessment of PIU has evolved from traditional psychometrics to supervised machine learning with multimodal data. This section analyzes the recent literature focused on predicting the impairment index using the HBN dataset from Kaggle.

A. Thematic Review of Specialized Literature

1) Tree Ensemble Models and Ordinal Optimization: Tree ensemble algorithms constitute the predominant approach for clinical tabular data. Niteesh and Pooja (2024) [5] evaluated XGBoost, LightGBM, CatBoost, and ExtraTrees on the CMI Kaggle data [4], demonstrating that addressing the problem through continuous regression and threshold tuning to maximize QWK surpasses nominal classification, achieving a QWK of 0.720 with Bayesian optimization.

Malallah and Iqbal (2026) [3] evaluated Gradient Boosting Regressor (GBR), LASSO, and Naive Bayes on 2,736 labeled records, achieving an accuracy of 84.5% and explaining the importance of attributes using SHAP values (PCIAT, BIA, and FitnessGram). For his part, Domingos (2025) [6] confirmed the effectiveness of XGBoost and decision trees by integrating passive actigraphy measurements and physical biometrics.

2) Multimodal Architectures, Advanced Imputation, and Neural Networks: The MREM-IUP (2024) model developed a multi-regressor ensemble (LightGBM, XGBoost, CatBoost, and TabNet) that incorporated Autoencoders to compress accelerometry data and used `KNNImputer` for the reconstruction of clinical blocks, achieving an outstanding QWK of 0.920 [2].

Wang et al. (2025) [7] proposed a multimodal cascade framework processing continuous accelerometry with 1D Convolutional Neural Networks (CNN-1D) and combining them with a tabular LightGBM model (AUROC $= 0.88$). Finally, Jović et al. (2024) [8] evaluated Artificial Neural Networks (ANN), Genetic Algorithms, and Random Forest on 2,113 participants, highlighting the need for feature selection (LASSO and ElasticNet) due to high dimensionality.

B. Identification of Research Gaps

The critical analysis of previous works reveals three key methodological limitations:

1) **Absence of rigorous statistical validation:** lack of estimation of confidence intervals or non-parametric hypothesis tests to verify statistical significance between algorithms.
2) **Risk of information leakage in imputation:** application of global multivariate imputation prior to data partitioning in some studies.
3) **Poor treatment of imbalance in the severe class:** models optimized solely for global QWK tend to neglect sensitivity in the class *sii* $= 3$ (1.24% of the sample), requiring auditing with the Macro F1-Score.

C. Justification and Contribution of the Present Proposal

To address these gaps, our project proposes a methodological pipeline that:

- compares five different families of ML algorithms (parametric, KNN, tree ensembles, SVM, and neural networks);
- encapsulates standardization and imputation (`KNNImputer`) within each fold of a 5-Fold Stratified Cross-Validation, eliminating information leakage;
- incorporates the Macro F1-Score as a mandatory complementary metric to the QWK;
- evaluates dimensionality reduction quantitatively with PCA and UMAP.

Table II
TECHNICAL COMPARISON OF WORKS IN THE MACHINE LEARNING LITERATURE FOR THE PREDICTION OF PIU

Author(s) and Year	Problem Setup	Evaluated Algorithms	Validation	Metrics	Highlighted Results
Niteesh & Pooja (2024) [5]	Supervised / Ordinal classification (4 levels)	XGBoost, LightGBM, CatBoost, ExtraTrees, Bayesian optimization	Stratified 5-Fold CV	QWK, F1-Macro, Accuracy	QWK of 0.720 with XGBoost and Bayesian threshold adjustment
MREM-IUP (2024) [2]	Supervised / Multi-regressor ensemble	LightGBM, XGBoost, CatBoost, TabNet, Autoencoders, KNNImputer	Stratified 10-Fold CV	QWK, RMSE, MAE	QWK of 0.920 integrating actigraphy and BIA
Malallah & Iqbal (2026) [3]	Supervised / Multiclass classification	GBR, LSTM, CNN-1D, LASSO, Naive Bayes	Stratified 10-Fold CV	Accuracy, F1-Score, QWK	GBR achieved 84.5% accuracy and interpretability with SHAP
Wang et al. (2025) [7]	Supervised / Multimodal cascade	XGBoost, LightGBM, Random Forest, CNN-1D	Stratified 5-Fold CV	AUROC, QWK	AUROC of 0.88 combining sensors with tabular data
Jović et al. (2024) [8]	Supervised / Ordinal classification	Random Forest, Gradient Boosting, ANN, Genetic Algorithms, LASSO	Stratified 10-Fold CV	Accuracy, F1-Score, Importance	Random Forest and GBR achieved the best F1-Macro (0.623)

D. Comparative Table of Reference Works

Table II summarizes the main works of the specialized literature evaluated for the formulation of the state of the art.

REFERENCES

[1] S. Sohn *et al.*, "The association between smartphone addiction and sleep, depression, anxiety, and stress in children and adolescents: a systematic review and meta-analysis," *Frontiers in Psychiatry*, vol. 10, p. 921, 2019.

[2] M. Karthiga, R. Sureshraja, A. Archudha, N. Kavya D, B. Deepshree, and A. K. Tripathy, "MREM-IUP – A MultiRegressor Based Ensemble Model for Assessing the Internet Addiction in Youth Using Physical and Behavioural Indicators," *South European Journal of Public Health (SEEJPH)*, vol. 25, pp. 1182–1189, Oct. 2024.

[3] F. L. Malallah and K. Iqbal, "Predicting Signs of Problematic Internet Use to Analyze Children's Health Using Machine Learning," in *Proc. 19th Int. Joint Conf. on Biomedical Engineering Systems and Technologies (BIOSTEC – BIOSIGNALS)*, vol. 1, 2026, pp. 247–254.

[4] Child Mind Institute and Kaggle, "Child Mind Institute — Problematic Internet Use: Relating Physical Activity to Problematic Internet Use," Kaggle Competition, 2024. [Online]. Available: https://www.kaggle.com/competitions/child-mind-institute-problematic-internet-use

[5] N. K. R and T. S. Pooja, "Predicting Problematic Internet Use in Children Using Feature-Rich Structured Data with Ensemble Machine Learning and Bayesian Optimisation," in *Proc. Asia-Pacific Signal and Information Processing Association Annual Summit and Conf. (APSIPA ASC)*, 2025, pp. 1627–1632.

[6] E. S. S. Domingos, "Automação Inteligente para Predição do Uso Problemático da Internet em Jovens com Base em Indicadores Físicos e Aprendizado de Máquina," TCC em Engenharia Mecatrônica, Universidade Federal do Rio Grande do Norte (UFRN), Natal, Brasil, 2025.

[7] S. Wang, W. Pan, and C. Su, "Identification of Problematic Internet Use with Machine Learning," Horace Greeley High School & Weill Cornell Medicine, Tech. Rep., 2025.

[8] J. Jović, A. Ćorac, A. Stanimirović, M. Nikolić, M. Stojanović, Z. Bukumirić, and D. Ignjatović Ristić, "Using machine learning algorithms and techniques for defining the impact of affective temperament types on problematic internet use in adolescents," *Frontiers in Public Health*, vol. 12, Art. no. 1326178, May 2024.